

1/10/2026 - CONCLUSION ON DUAL-MODE CONTROL + FIRST CASE STUDY

∞ -HORIZON CONST. CONTROL

$\longleftrightarrow$ FINITE-HORIZON CONST. CONTROL.

$$\min \left\{ \sum_{k=0}^{\infty} \ell(u_k, x_k) \right\}$$

s.t. $V(x_0)$

$$\begin{cases} x_0 \text{ known} \\ x_{k+1} = f(x_k, u_k), k=0, 1, \dots \end{cases}$$

$$\begin{cases} u_k \in U, k=0, 1, \dots \\ x_k \in X \end{cases}$$

$$\min_{\{u_0, u_1, \dots, u_{N-1}\}} \left\{ \sum_{k=0}^{N-1} \ell(u_k, x_k) + \underline{V(x_N)} \right\}$$

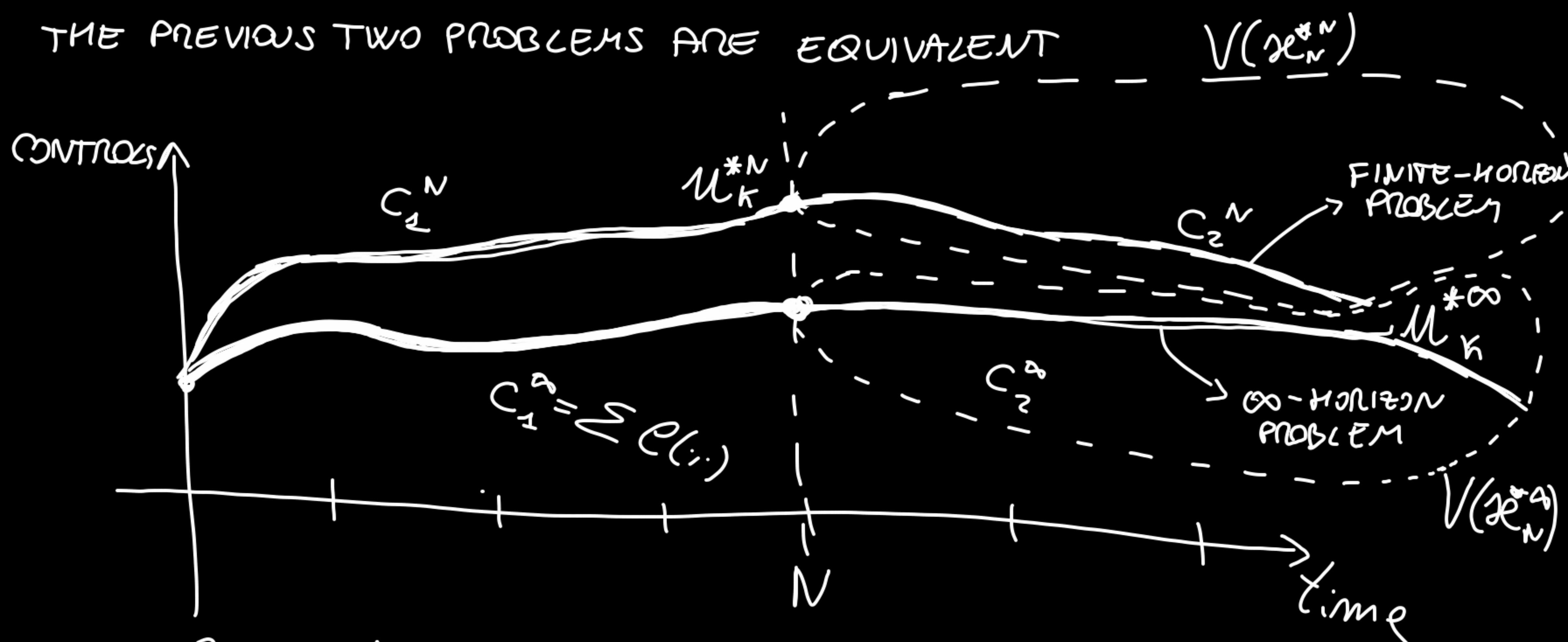
s.t.

$$\begin{cases} x_0 \text{ known} \\ x_{k+1} = f(x_k, u_k), k=0, 1, \dots, N-1 \end{cases}$$

$$\begin{cases} u_k \in U, k=0, 1, \dots, N-1 \\ x_k \in X, k=0, 1, \dots, N \end{cases}$$

$V(x): \mathbb{R}^m \rightarrow \mathbb{R}$ - COST OF OPTIMALLY CONTROLLING THE SYSTEM STARTING FROM STATE x . ($V(x) = \infty$ IF PROBLEM FROM x IS NOT FEASIBLE)

THE PREVIOUS TWO PROBLEMS ARE EQUIVALENT



$$C_1^N + C_2^N < C_1^\infty + C_2^\infty$$

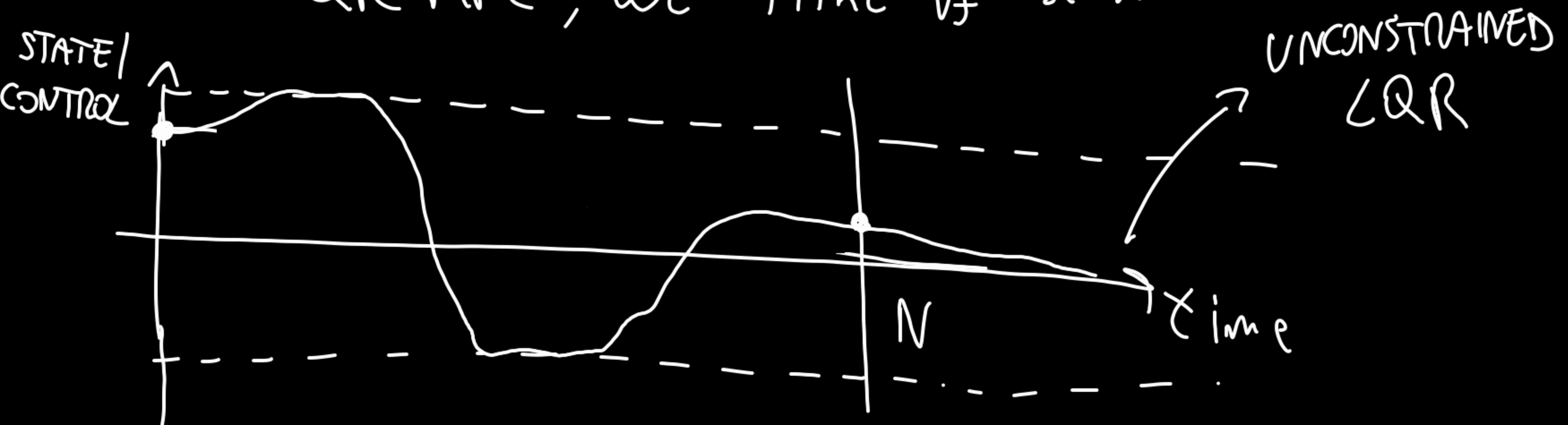
PROBLEM

$V(x)$ IS NOT KNOWN IN GENERAL.

EXCEPTION: UNCONSTRAINED LQR

$$V(x) = x^T W x$$

SO, IN LQR MPC, WE TAKE $V_f = x^T W x$



LQR MPC FORMULATION

$$\mathcal{P}^N(x_k): \min \left\{ V^N(x_k) = \sum_{i=0}^{N-1} \left(x_{i|k}^T Q x_{i|k} + u_{i|k}^T R u_{i|k} \right) + V_f(x_{N|k}) \right\}$$

s.t.

$$\begin{cases} x_{0|k} = x_k \\ x_{i+1|k} = A x_{i|k} + B u_{i|k}, \quad i=0, 1, \dots, N-1 \end{cases}$$

$$\begin{cases} u_{i|k} \in U, \quad i=0, 1, \dots, N-1 \\ x_{i|k} \in X, \quad i=0, 1, \dots, N \end{cases}$$

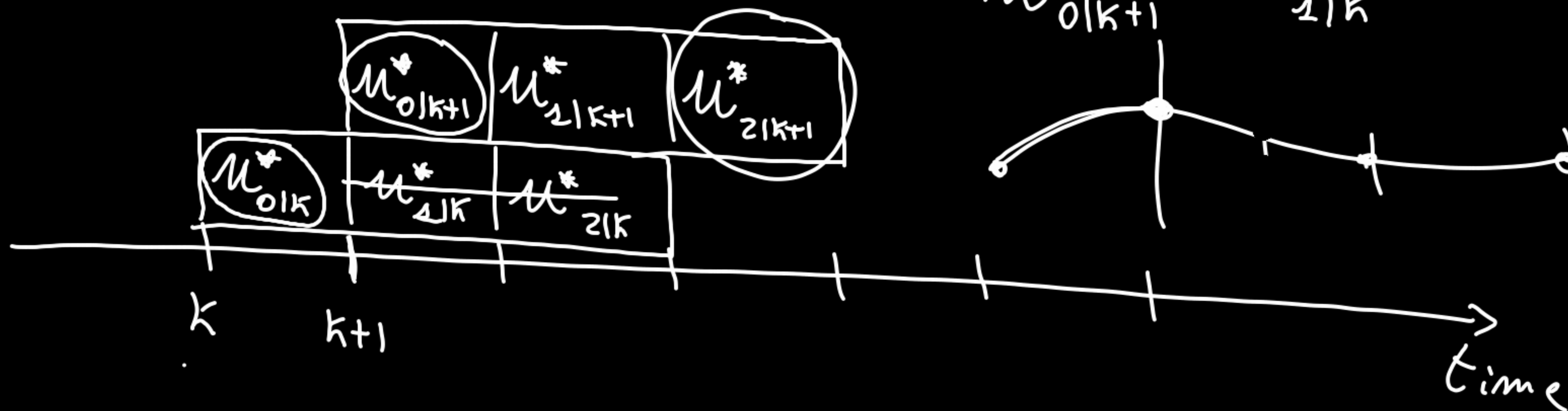
WITH:

$$V_f(x) = x^T W x$$

N LARGE ENOUGH.

(∞ -HORIZON CONSTRAINED LQR)

- WE USE MPC WITH PROPER V_f, W .
- $N=3$



$$u_{0|k+1}^* = u_{1|k}^*$$

$$x_{k+1} = Ax_k + Bu_{0|k}^* = \underline{f(x_k, u_{0|k}^*)}$$

(NOMINAL ASSUMPTION / PERFECT MODEL, NO NOISE / DISTURBANCE)

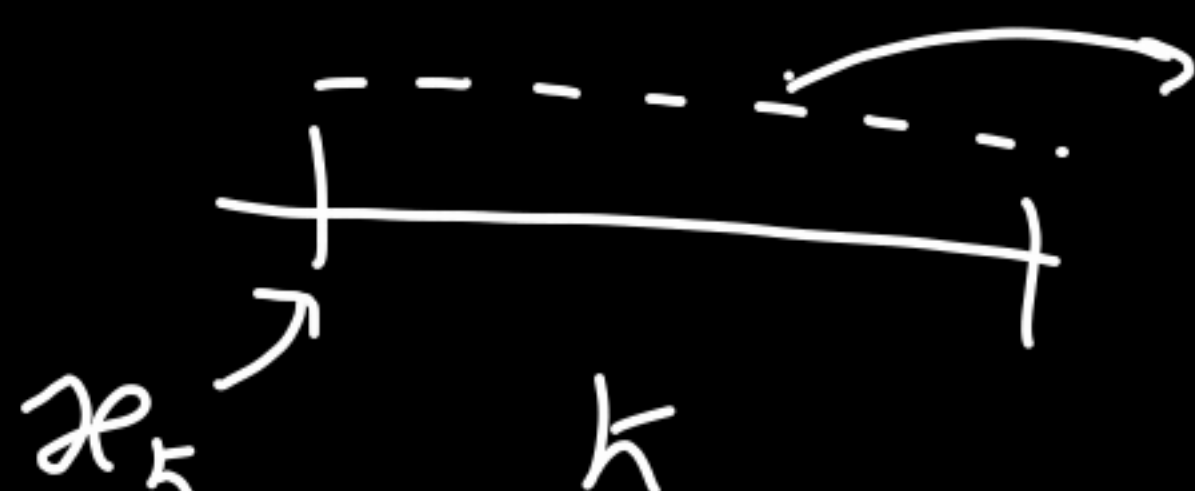
IN REALITY

- MODEL IS NOT PERFECT
- NOISE
- DISTURBANCES

$$u_{0|k+1}^* \neq u_{1|k}^*$$

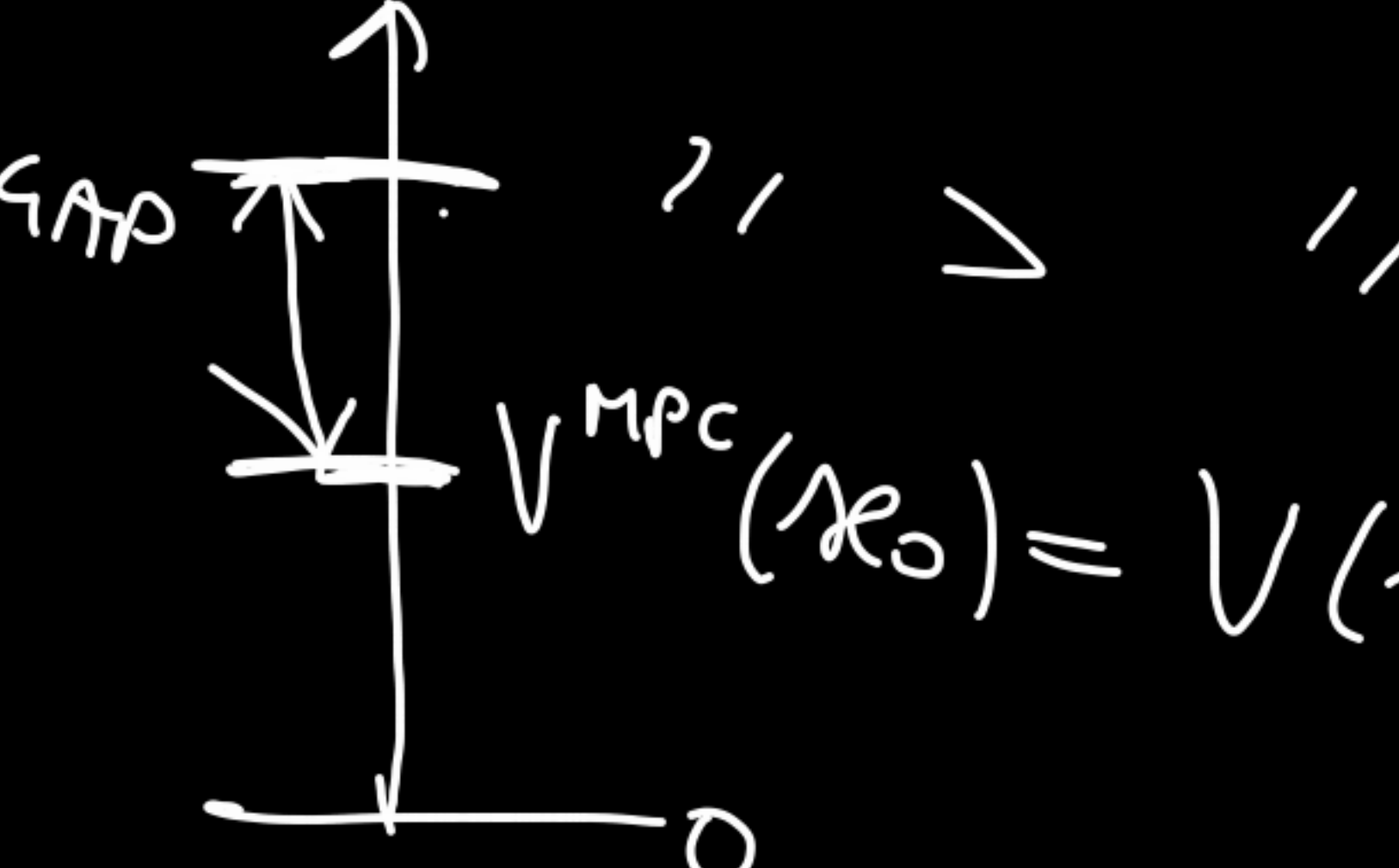
HOW TO COMPUTE COST OF MPC CONTROL

STAGE COST $\ell(u_k, x_k)$



$$V^{\text{MPC}}(x_0) = \sum_{k=0}^{\infty} \ell(x_k, u_{0|k}^*)$$

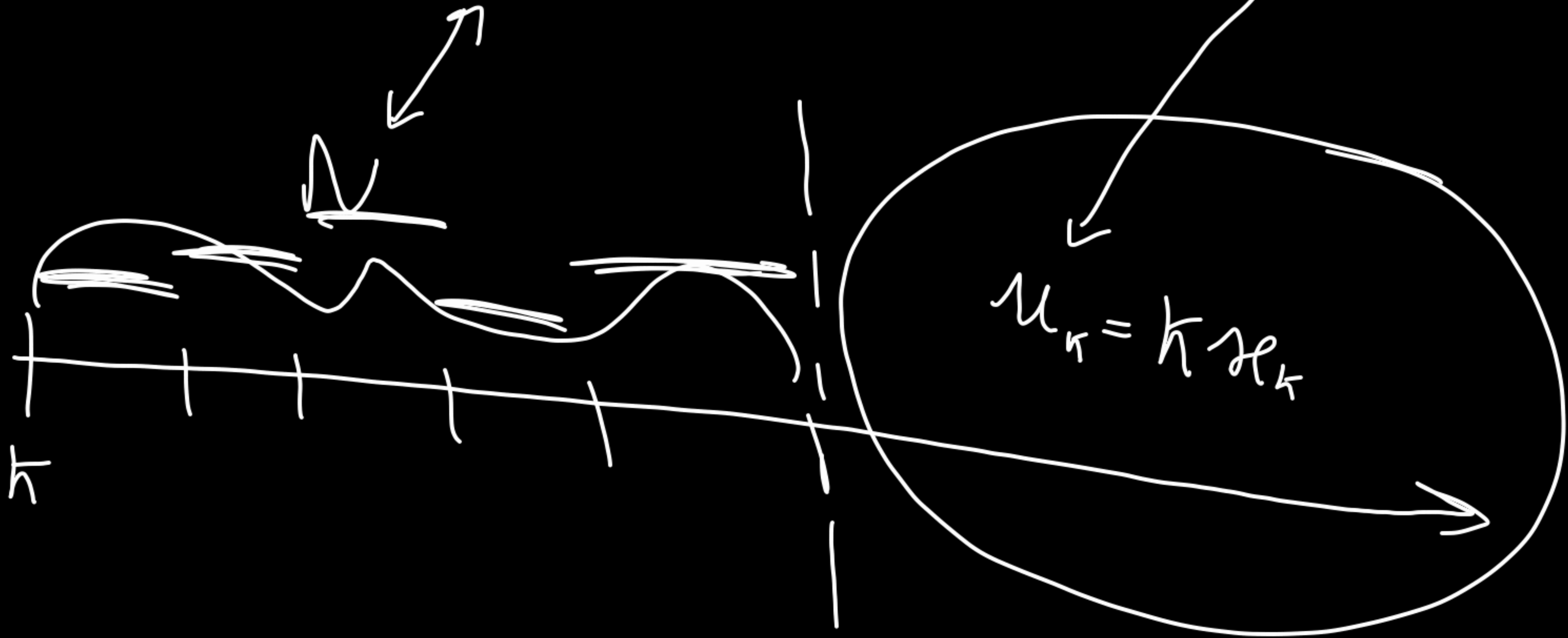
$$x_{k+1} = f(x_k, u_{0|k}^*)$$



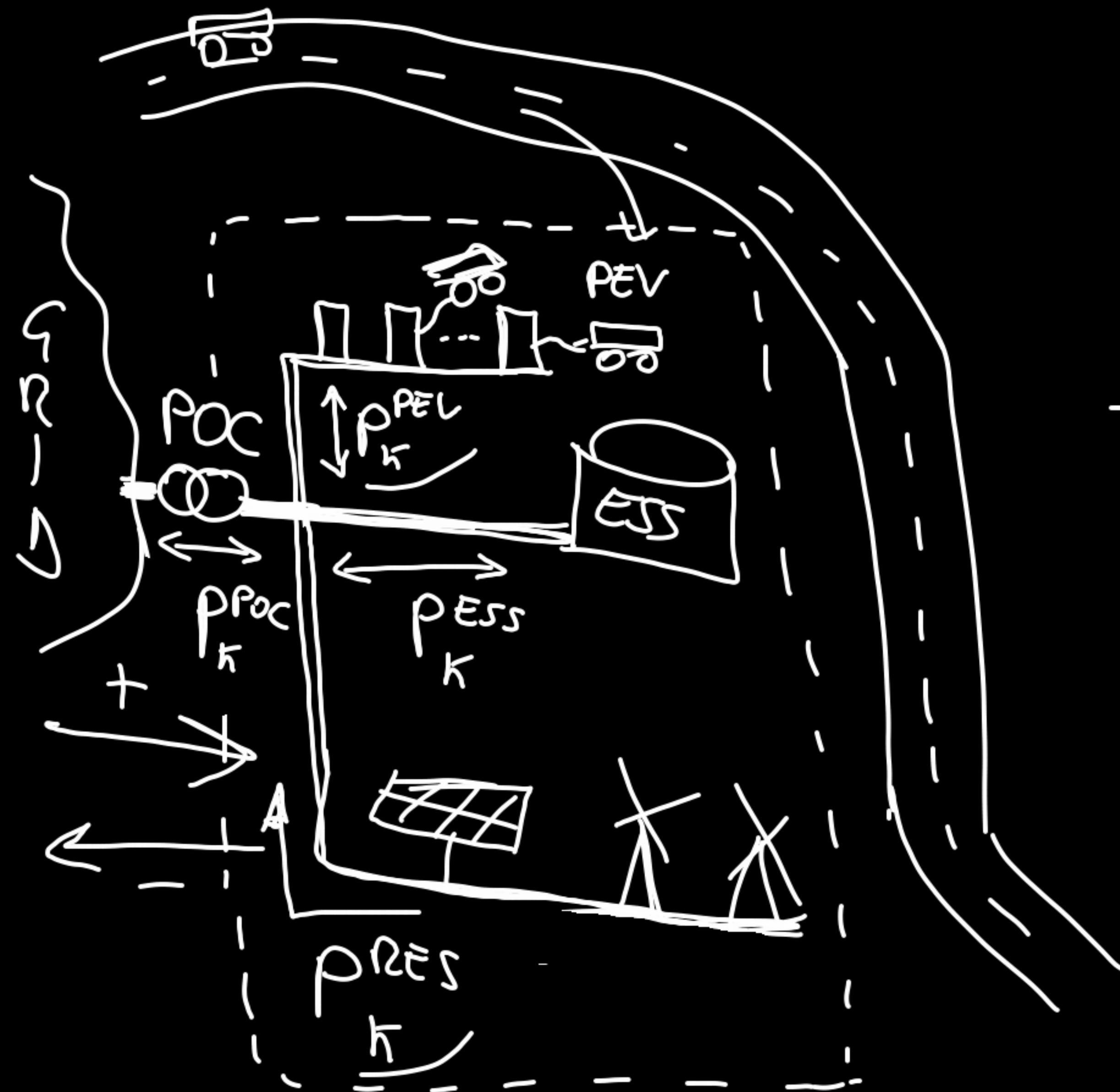
$V^{\text{MPC}}(x_0) = V(x_0)$ IN IDEAL CONDITIONS, MPC IS OPTIMAL

DUAL-MODE CONTROL

$$\sum_{i=0}^{N-1} \left(x_{ik}^T Q x_{ik} + u_{ik}^T R u_{ik} \right) + \underbrace{x_{Nk}^T V x_{Nk}}$$

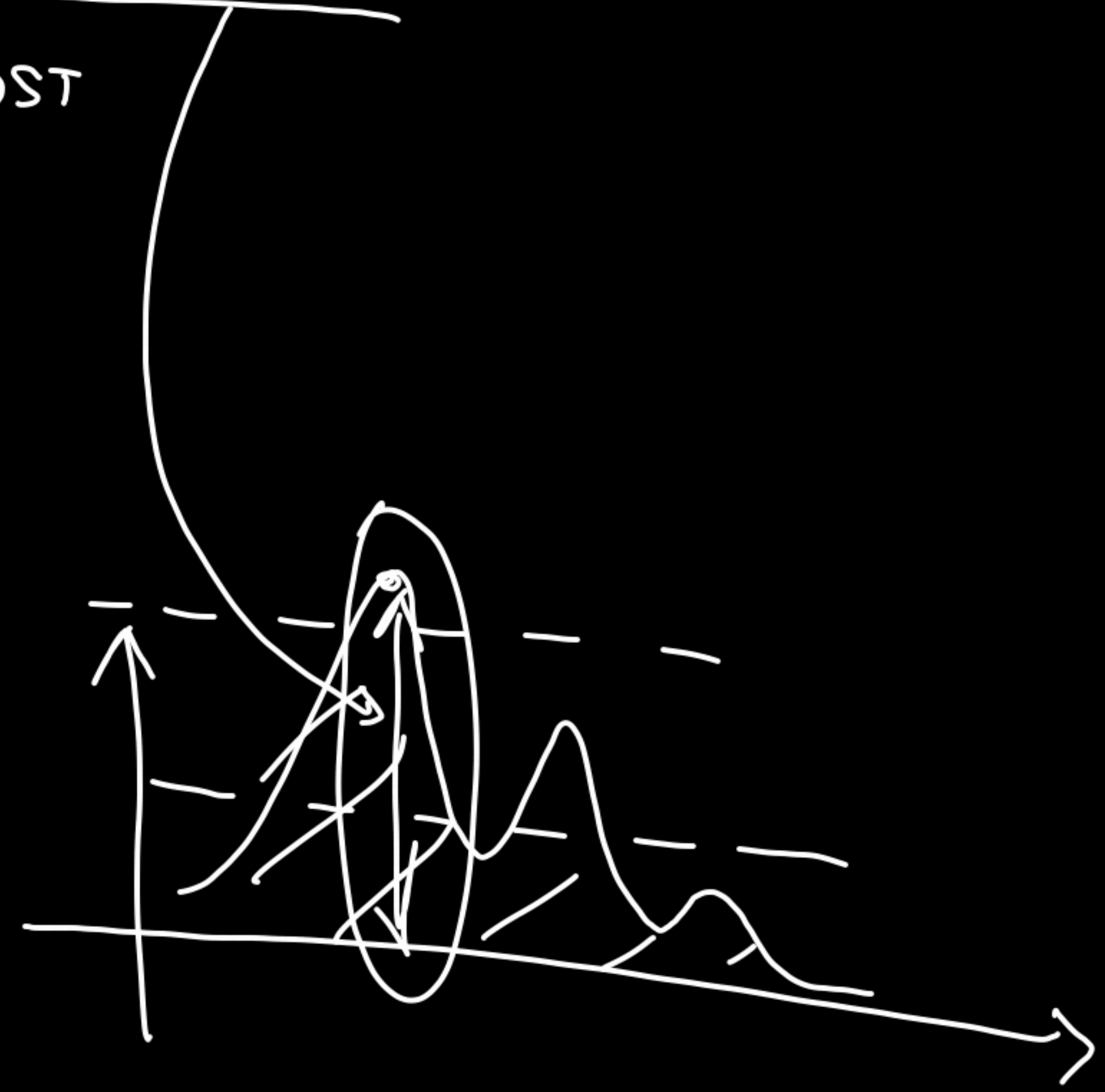
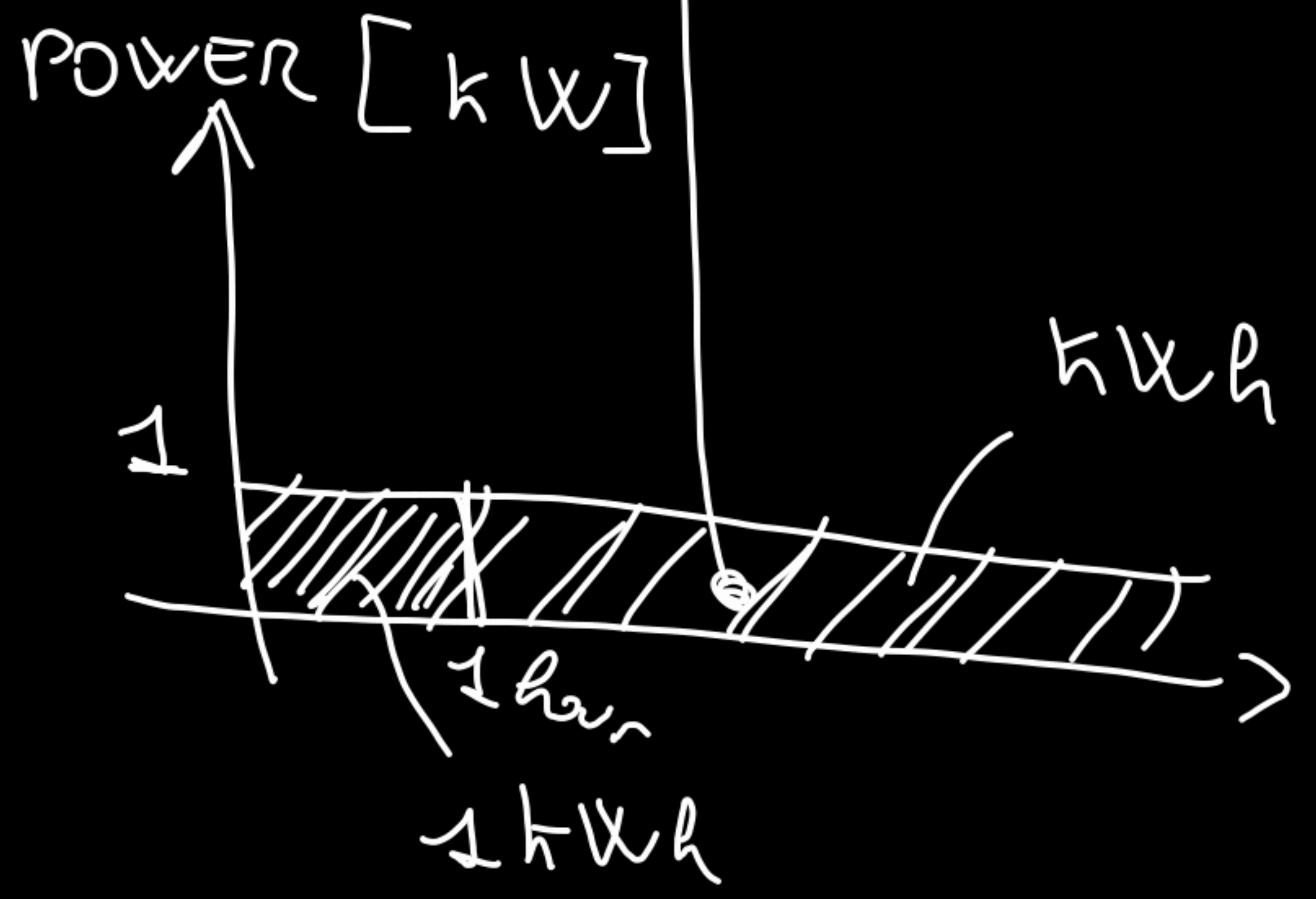
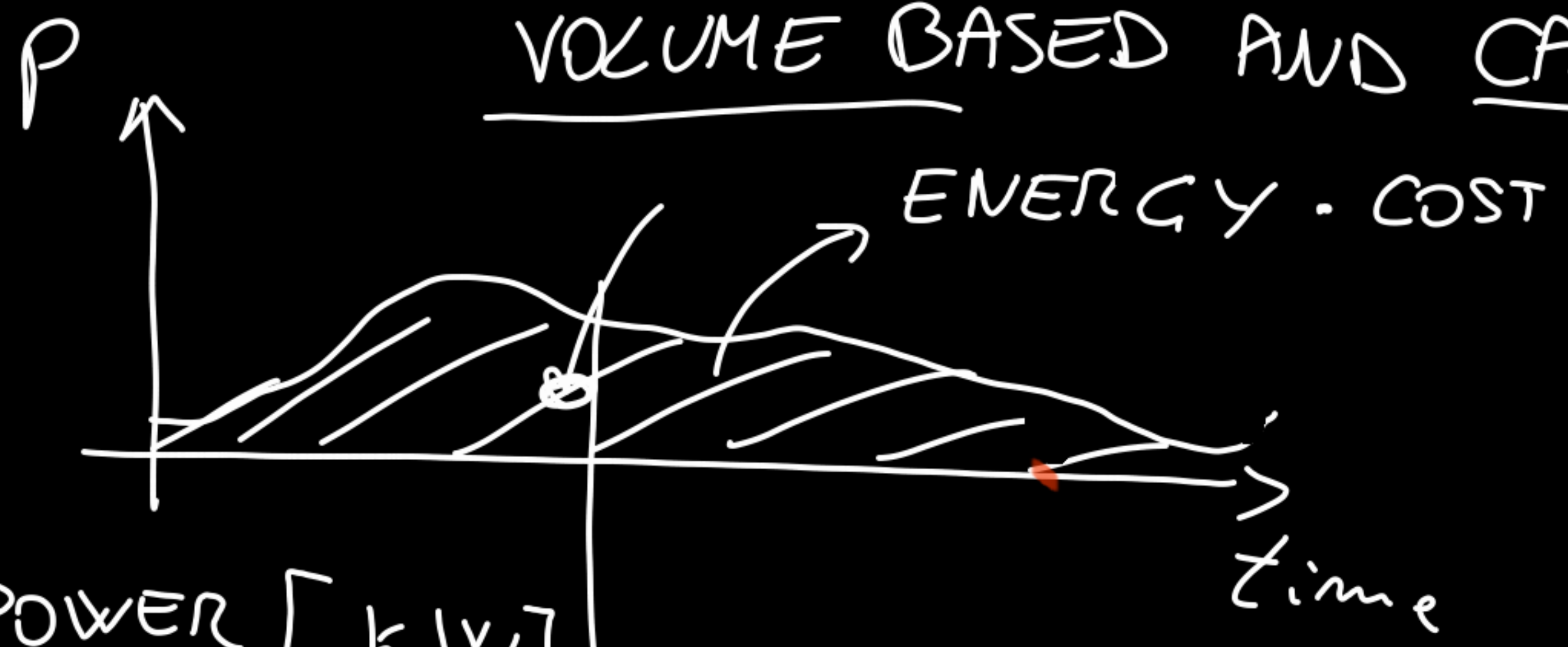


FIRST PRACTICAL CASE STUDY - CONTROL OF A CHARGING AREA



- PEV ARRIVE AND WANT TO RECHARGE BATTERIES AS FAST AS POSSIBLE (MAX. POWER)
~ 100' kW
- PROBLEM → EVEN WITH FEW CLIENTS, WE REACH HUGE POWER DEMAND → VERY HIGH FLOW AT POC
 - VERY EXPENSIVE HARDWARE AT POC (TRANSFORMER, CABLES)
 - HIGHER ELECTRICITY BILL

VOLUME BASED AND CAPACITY BASED TARIFFS



GOALS:

① RECHARGE AS FAST AS POSSIBLE — END-USER GOAL

② $\left[\begin{array}{l} \text{KEEP } P_k^{POC} \text{ AS CLOSE AS POSSIBLE TO } 0. \\ \text{(OR, AS LOW IN MAGNITUDE AS POSSIBLE)} \end{array} \right]$

MPC FORMULATION:

VARIABLES.

CONTROL VARIABLES: $P_k^{ESS} [kW]$

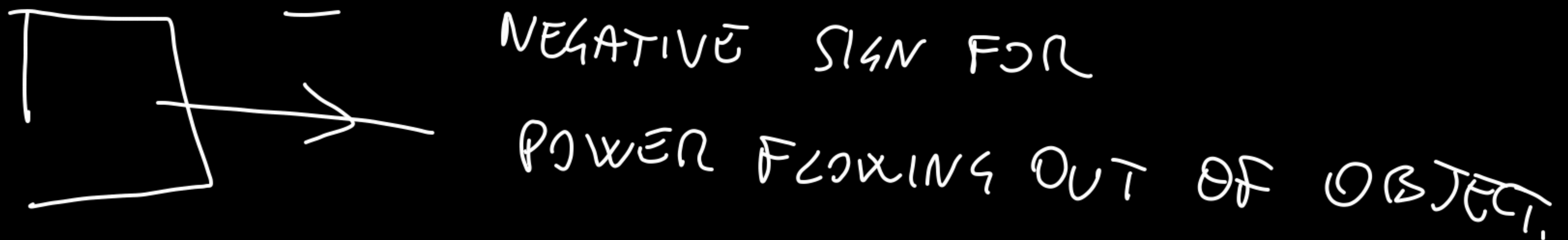
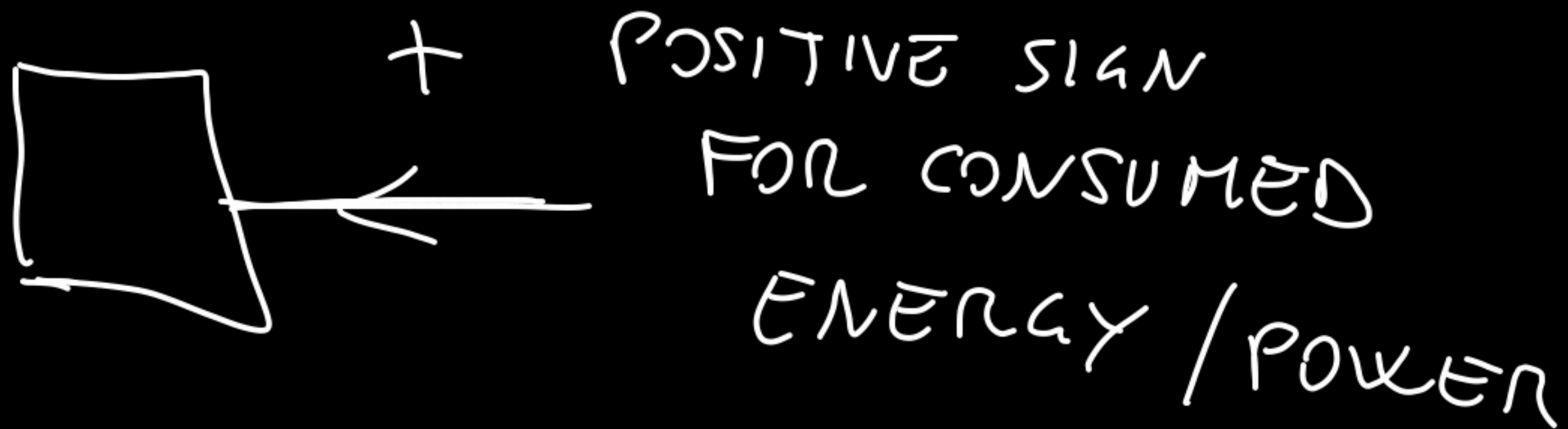
STATE VARIABLE: $x_k^{ESS} [kWh]$ ENERGY IN THE
ESS AT TIME k .

CONTROLLED VARIABLE: $P_k^{Poc} [kW]$

DISTURBANCES: $P_k^{REV} [kW], P_k^{RES} [kW]$ - I HAVE PERFECT
PREDICTION OF THEM

POWER SIGN CONVENTION

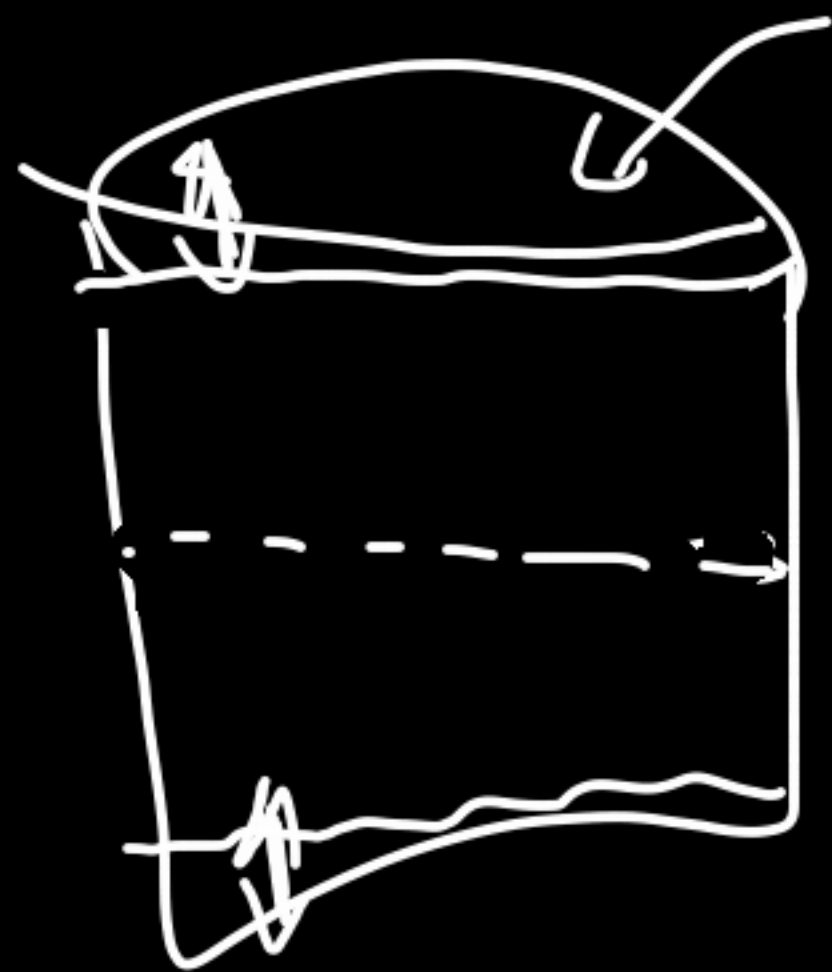
- CONSUMER CONVENTION



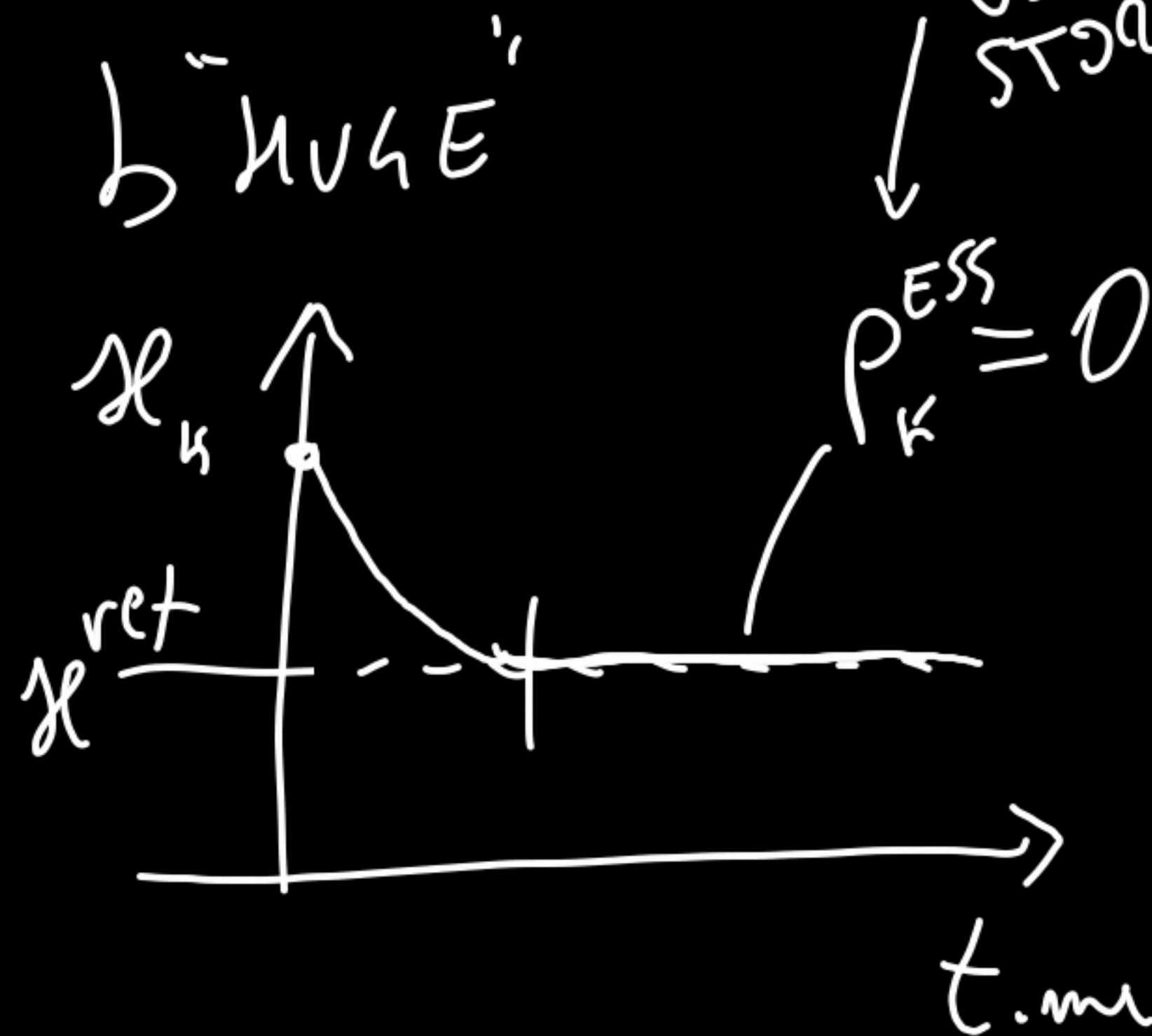
MPC FORMULATION

$$\min \sum_{i=0}^{N-1} \left\{ a \cdot (p_{i|k}^{pac})^2 + b (x_{i|k} - x^{ret})^2 \right\} + V_f(x_{N|k})$$

b SHOULD BE SMALL COMPARED TO a.



"NICE TO HAVE"



CONSTRAINTS

GLOBAL CONSTRAINT — POWER BALANCE EQUATION
(LINKS ALL POWERS)

$$\underline{P_i^{POC}} = P_{i|k}^{PEV} + \underline{P_{i|k}^{ESS}} - P_{i|k}^{RES}, i=0,1,\dots,N$$

